

Avery Martin¹, Dr. Mengdi Huai², Dr. Peng Xu¹

Machine Learning-Enhanced Ab Initio Force Field for Intermolecular Interactions

Introduction

- Intermolecular forces are weak, long-range interactions that arise primarily from electrostatic effects between molecular charge distributions induced during molecular interactions.
- One example is dispersion energy, which is weak attraction due to the correlated movement of electrons.

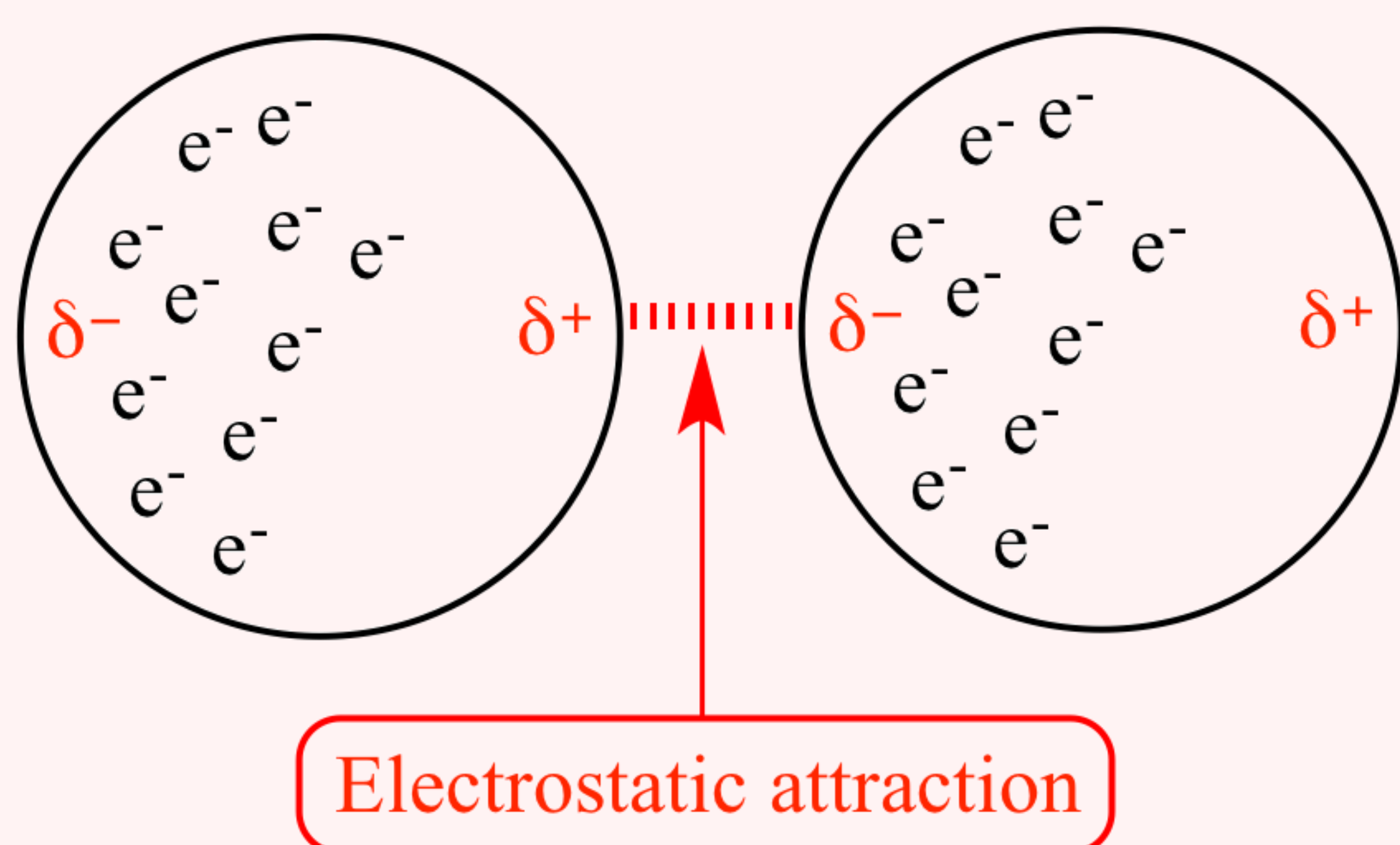


Figure 1. A diagram of dispersion interactions

- A variety of computational methods exist for calculating intermolecular interaction energies. However, these methods often involve tradeoffs—some offer high accuracy at the cost of significant computational power, while others prioritize speed but compromise precision.
- The Effective Fragment Potential (EFP) method is an *ab initio* force field that computes intermolecular interactions. It is derived from quantum mechanics with no empirically fitted parameters. EFP delivers interaction energies comparable to second-order perturbation theory, but at a fraction of the computational cost.

Discussion and Future Work

- The main next step would be to apply the data I created to other systems of molecules. The applications of having a method of predicting interaction energies that's both low in computational cost and highly precise are vast. A high priority would be metal-protein interactions, since metal-protein interactions are highly important to drug engineering and our understanding of certain disorders.
- Another next step would be to produce substantially more interaction energy data for the ML model to be trained on. The model is a neural network, and neural networks tend to need a very large set of data to have high accuracy and precision.

References

- (1) *Illustrated Glossary of Organic Chemistry - London Force*. www.chem.ucla.edu. https://www.chem.ucla.edu/~harding/IGOC/L/london_force.html
- (2) Barca, G. M. J.; Bertoni, C.; Carrington, L.; Datta, D.; De Silva, N.; Deustua, J. E.; Fedorov, D. G.; Gour, J. R.; Gunina, A. O.; Guidez, E.; Harville, T.; Irle, S.; Ivanic, J.; Kowalski, K.; Leang, S. S.; Li, H.; Li, W.; Lutz, J. J.; Magoulas, I.; Mato, J. Recent Developments in the General Atomic and Molecular Electronic Structure System. *The Journal of Chemical Physics* **2020**, 152 (15), 154102. <https://doi.org/10.1063/5.0005188>.
- (3) Xu, P.; Westheimer, B. M.; Schlinsog, M.; Tosaporn Sattasathuchana; Elliott, G.; Gordon, M. S.; Guidez, E. The Effective Fragment Potential: An Ab Initio Force Field. *Elsevier eBooks* **2023**, 1 (1), 153–161. <https://doi.org/10.1016/b978-0-12-821978-2.00141-0>.

Objectives

- Produce large amounts of interaction energy data for the computer science team to train an ML model with.
- Create an optimized database of amino acid conformer coordinates for further applications to systems of proteins.

Methods

- I used GAMESS (General Atomic and Molecular Electronic Structure System) to calculate the total interaction energies of a large database of molecules. GAMESS is used for a wide range of quantum chemistry calculations.
- Specifically, I used the EFP (Effective Fragment Potential) method to compute the interaction energies. EFP breaks down large systems into fragments before calculating the different types of energy to save computing power.
- Jobs were submitted on the NOVA cluster from ISU using the SLURM job scheduler to organize the submitted jobs.
- Code modification was made to output S_{ij} and R_{ij} values for each calculation. These variables are part of the equation for penetration energy. The computer science side used these numbers to train the machine learning models.

$$E_{e-e}^{pen} = -2\left(\frac{1}{-2\ln|S_{ij}|}\right)^{\frac{1}{2}} \frac{S_{ij}^2}{R_{ij}}$$

S_{ij} : Intermolecular overlap integral between LMOs i and j
 R_{ij} : The distance between centroids of LMOs i and j

Figure 2. The equation for penetration energy. S_{ij} and R_{ij} are defined below the equation.

Results

FRAGMENT-FRAGMENT INTERACTION ENERGIES		FREE ENERGY IN SOLVENT: -202646.35 KCAL/MOL	
CHARGE-CHARGE	= -0.0023757819	INTERNAL ENERGY IN SOLVENT: -202636.62 KCAL/MOL	
CHARGE-DIPOLE	= -0.0043063981	DELTA INTERNAL ENERGY: 0.00 KCAL/MOL	
CHARGE-QUADRUPOLE	= -0.0047534661	ELECTROSTATIC INTERACTION: -9.73 KCAL/MOL	
CHARGE-OCTUPOLE	= 0.0002179785	PIEROTTI CAVITATION ENERGY: 0.00 KCAL/MOL	
DIPOLE-DIPOLE	= -0.0008127373	DISPERSION FREE ENERGY: 0.00 KCAL/MOL	
DIPOLE-QUADRUPOLE	= -0.0022544413	REPULSION FREE ENERGY: 0.00 KCAL/MOL	
QUADRUPOLE-QUADRUPOLE	= -0.0014311137	TOTAL INTERACTION: -9.73 KCAL/MOL	
OVERLAP PEN. ENERGY	= -0.0021016063	TOTAL FREE ENERGY IN SOLVENT: -202646.35 KCAL/MOL	
ELECTROSTATIC ENERGY = -0.0178175663			
REPULSION ENERGY = 0.0153321026			
POLARIZATION ENERGY = -0.0042998917			
TOTAL DISPERSION ENERGY (E6+E7+E8) = -0.0025801538			
E7 DISPERSION ENERGY = 0.0015726003			
E6 DISPERSION ENERGY = -0.0023665564			
E8 DISPERSION ENERGY = -0.0017861977			
CHARGE TRANSFER ENRGY = -0.0012615909			
FINAL EFP ENERGY = -0.0106271000			

Figure 3. Sample outputs from the amino acid database (left) and EFP calculations (right)

- I successfully created a database with a total of 470 EFP calculations from a wide database of molecular systems. This data was used by the computer science team to make an ML model to predict interaction energies.
- I was also successful in creating a database of optimized coordinates for every amino acid conformer, which totaled to 168 coordinate files. These files will be used for future protein calculations.